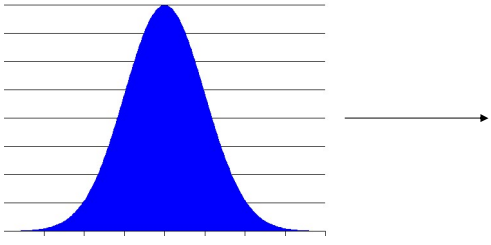
Why Statistics?

- Deal with uncertainty in repeated scientific measurements.
- Draw reliable conclusions from data.
- Design valid experiments and draw reliable conclusions.
- Be a well-informed member of society.

1

Population and statistical inferences

A population is the entire group of objects about which you want information.



Population is completely identified: Both the shape and the parameters are known.

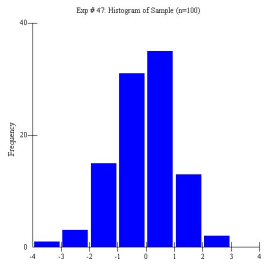
Compute probabilities, expectations, etc., for random variables X from this population.
X : future observations from this population

Deductive inferences (Tumdengelim)

2

Sample and statistical inferences

A sample is a subset of observations selected from the population. It is used to gain knowledge about the population.



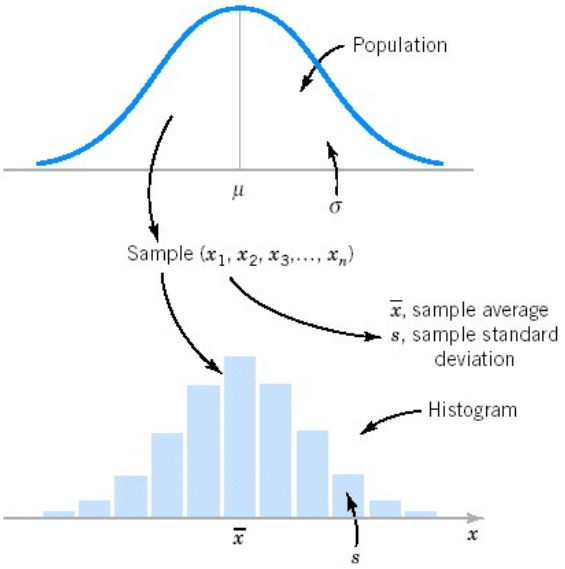
Estimate the unknown parameters of this population.

The shape of the population is known, but not its parameters.

Inductive inferences (Tumevarim)

3

Relationship between a population and a sample



4

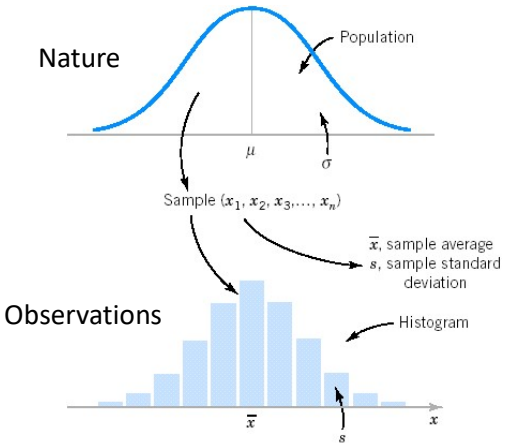
Games online?

- Have you played Tavla online?
- How do you think the dice are rolled?

- Can we also simulate a die roll? → Labs!

5

How to link the two worlds? Population and Sample



- If we can play both sides, we can understand the link between them better!
- Simulations where you can generate die rolls, coin tosses and more → representing the God who controls the Nature
- Analyzing the data that you yourself generated will show you how you can use the data to understand the population

6

In the Labs:

- You will play the God sometimes, and yourselves in other times…
- God: Simulations (many data sets can be generated)
- Yourself: Statistical analysis via Python (just one data set is there to analyze)

7

In the lectures:

- You will:
  - understand the link between the two worlds mathematically,
  - learn which questions to ask and how to answer them when you have a problem and a corresponding data set.

8

Example 1 Quality Control

- 1000 rods are produced to meet the specification 0.45 ± 0.02 cm
- No time and no resource to check all 1000 rods.
- What to do?

Here are some relevant questions:

1. How large is a typical difference between the sample proportion and the population proportion for this kind of sample?
2. What interval gives a good estimate of the percentage of acceptable rods in the population with reasonable certainty?
3. How certain can the engineer be that at least 90% of the rods are good?

Statistics can help us to address questions like these.

9

Sampling

- A **population**: entire collection of objects or outcomes about which information is sought.
- A **sample**: a subset of a population, containing the objects or outcomes that are actually observed.
- A **simple random sample (SRS)** of size *n* : a sample chosen by a method in which each collection of *n* population items is equally likely to comprise the sample, just as in the lottery.

OR

- The random variables $X_1,\ldots,X_n$ are said to **simple random sample** of size *n* if:
  - (1) X_i's are independent r.v.s
  - (2) X_i's have the same probability distribution

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Independence

- The items in a sample are said to be **independent** if knowing the values of some of the items does not help to predict the values of the others.

  → choosing one item for the sample has no effect on the choice of other items

- Items in a simple random sample may be treated as independent in most cases encountered in practice, unless the population is finite, and the sample comprises a substantial fraction (more than 5%) of the population.

11

How to draw a random sample?

- A professor's question: what is the average height of students at a university?
- Population?
- How to choose a sample size of 100?

12

A sample of convenience

A sample that is obtained in some convenient way and is not drawn by a well-defined random method.

Things to consider:

- Convenience samples may differ **systematically** in some way from the population.
- Thus, they should only be used when it is not feasible to draw a random sample.
- If you do have a convenience sample, think carefully about the ways in which the sample might differ systematically from the population.

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How to draw a sample of convenience?

- A professor's question: average height of students at a university?
- Population?
- How to choose a sample of size of 100?

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